

RESEARCH ARTICLE

Robotization of Miniature-Scale Radio-Controlled Excavator: A New Medium for Construction-Specific DNN Data Generation

SEYEDEH FATEMEH SAFFARI¹, DAEHO KIM¹, AND BYUNGJOO CHOI²

¹Department of Civil and Mineral Engineering, University of Toronto, Toronto, ON M5S 1A4, Canada

²Department of Architectural Engineering, Ajou University, Suwon-si, Gyeonggi-do 16499, Republic of Korea

Corresponding author: Daeho Kim (civdaeho.kim@utoronto.ca)

This work was supported in part by the Natural Sciences and Engineering Research Council of Canada (NSERC) Awards through the Discovery Grant Funded by the Development and Evaluation of Real-Like Synthetic Construction Images to Enhance the Performances of Deep Neural Network for Construction Applications under Grant RGPIN-2022-04429; and in part by the Collaborative Research and Development Grants, Building Information Modeling (BIM)-Driven Productivity Improvements for the Canadian Construction Industry under Grant 530550-2018.

ABSTRACT Digital management of excavators has seen limited progress due to challenges in artificial intelligence (AI) training. The AI required for digital twinning of excavators necessitates a large volume of diverse imagery data, currently scarce in the construction domain. Moreover, the absence of deployable robot agents hinders reinforcement learning, impeding task-oriented AI development. In response, we introduce an innovative approach utilizing a miniature-scale, radio-controlled excavator (RC-excavator). This presents a cost-effective method for automated data collection and labeling, as well as interactive reinforcement learning. The RC-excavator's electric circuit was modified, its motion dynamics were modeled, and it was fully robotized for precise computer-directed motion control. Statistical validation of its motions achieved a Normalized Range Adjusted Accuracy (NRAA) of 99.14% for the bucket, 97.97% for the main arm, and 98.63% for the cabin. This confirms its adequacy for image labeling and task-oriented automation research.

INDEX TERMS Automated labeling, construction equipment, digital twinning, interactive reinforcement learning.

I. INTRODUCTION

Construction enterprises are increasingly recognizing the potential value of innovations driven by artificial intelligence (AI), such as digital twinning and robotic automation. With excavators playing a central role in field construction, the productivity [1] and safety [2], [3] of excavator operations have a considerable impact on the overall success of a typical construction project. There is a growing consensus that the automation of excavator operations through AI can result in significant advancements in field productivity [4] and can potentially boost overall project profitability [5], [6]. The implementation of a digital twin for excavator operations, enabling a shift towards AI management, would open the

door to real-time monitoring and follow-up interventions [7]. This, in turn, could substantially reduce the frequency and scale of necessary reworks. Yet, endeavors to develop AI solutions specifically tailored to excavators are still in their nascent stages, with the development of visual AI (e.g., for pose estimation) and task-oriented AI based on reinforcement learning both are lagging.

Two primary gaps are restricting the development of excavator-focused AI solutions:

A. RESEARCH GAP #1: THE LACK OF A ROBUST AND PRACTICAL METHOD FOR 2D/3D LABELING OF MOBILE EXCAVATOR IMAGES

The core technology for the digital twinning of ongoing excavator operations is a visual AI engine powered by a deep neural network (DNN), which is trained for 3D pose estimation [8]. The effectiveness of this computer vision model

The associate editor coordinating the review of this manuscript and approving it for publication was Tao Liu¹.

depends on supervised learning, which in turn necessitates a large volume of training data (i.e., images with 2D/3D pose labels) [9], [10]. However, the unstructured, hazardous, dynamic, and proprietary nature of construction sites creates challenges in data collection, labeling, and sharing. As a result, available data sets for training are orders of magnitude smaller than needed. This data scarcity is significantly hindering the advancement of digital twinning research focused on excavator operations [8], [11], which in turn is hindering the development of effective task-oriented AI.

The unique properties of an excavator create challenges in accurately capturing 2D/3D poses. An excavator is distinct from other construction equipment due to its large articulated arm that features a high degree of freedom [12]. Critical functions such as digital twinning, monitoring of proximity to nearby workers, and the evaluation of productivity cycles all require accurate 3D pose information (i.e., a 3D skeleton) [13], [14], [15].

Effectively capturing the pose of the highly articulated arm of an excavator requires a robust DNN pipeline. The prevailing methodology involves a two-stage approach: (i) initially, 2D pose estimation is performed through a 2D pose DNN (e.g., Mediapipe [16], RTMPose [17], and Detectron2 [18]) and (ii) subsequently, 3D pose estimation is undertaken using a 2D-to-3D pose lifting DNN (e.g., VideoPose3D [19], MixSTE [20], or MotionBERT [21]). It is worth noting that this general approach requires the ground truth of both 2D and 3D poses as a fundamental prerequisite (i.e., 2D/3D coordinates of each joint location of an excavator on a camera coordinate system) [4].

The availability of data sets containing suitable images with 2D/3D pose labels is a primary rate-limiter in current research. Although some research has focused on collecting 2D labels of construction equipment [22], the quantity and diversity of the data collected remains inadequate. No publicly accessible 3D benchmark dataset exists for construction equipment, as confirmed by a comprehensive review of the literature conducted by the research team. A lack of motion capture technologies (e.g., OptiTrack [23] or Vicon [24]) capable of scaling up to work with real-scale excavators has exacerbated this lack of suitable datasets.

In response to this data shortage, researchers have had to explore inventive methods to label the 3D pose of an excavator. One notable strategy involves the generation of synthetic datasets. By harnessing computer graphics, these datasets place 3D models with known poses in virtual environments [3]. This synthetic approach offers an alternative to A) the process of manually labeling images of real construction environments (requiring prohibitive investments of manual labour) and B) the potential inaccuracies of marker-based methods [25] while delivering precise spatial information on the construction equipment skeleton [13], [26].

Although synthetic datasets offer a certain value, they also come with inherent challenges. One of the most significant is the domain shift in variables, including lighting conditions and the textures of objects. This domain

inconsistency can adversely affect the performance of DNN models trained exclusively on synthetic data when deployed in real-world scenarios [27]. Though remedies such as augmenting the training dataset by merging with real images [8], [11] or enhancing the fidelity of virtual images via Generative Adversarial Networks (GAN) [27] have been pursued, recent studies have consistently found that models trained on synthetic data underperform those trained on real data [29]. Thus, real data remains critically important in achieving an optimally performant DNN, even when supplemented with synthetic data. This requirement for real data points directly to the need for a robust and practical method for 2D/3D labeling of mobile excavators, enabling the creation of datasets of the size and reliability needed for AI optimization. In contrast to the shortcomings of synthetic data generation and marker-based systems, our marker-less approach provides a robust solution for collecting ground-truth pose information, while circumventing the challenges posed by real-world construction environments.

B. RESEARCH GAP #2: THE LACK OF A SAFE AND PRACTICAL REAL AGENT FOR REINFORCEMENT LEARNING

Working in conjunction with visual AI, task-oriented AI governs a robot's decision-making and motion control for a specific task, which is central to excavator automation [30]. The development and optimization of task-oriented AI primarily employs reinforcement learning that depends on bi-directional responsive data that captures the continuous interaction of the agent with its surroundings [31]. While closed-loop predictive control of excavator tasks using radio-controlled excavators (RC-excavator) is effective, task-oriented AI can adapt to changing conditions through continuous learning [32]. Currently, the unique challenges in developing an effective live excavator agent (e.g., a robotized excavator) capable of seamlessly interacting with real surroundings is a second primary barrier to developing effective excavator-specific AI. The ability to employ reinforcement learning is highly constrained without a live agent capable of generating authentic bi-directional data [30].

The size of typical real-world excavators precludes their use as an effective live agent. Commercial excavators can weigh over 200 tons, with 40-ton excavators commonly used on construction sites. The full dimensions of a 40-ton excavator are approximately 32 feet in length, 10 feet in width, and 12 feet in height. The logistical complexities, space limitations, and safety concerns that accompany this size mean that real-scale excavators have proven impractical as live training agents [33].

The challenges to using real-scale construction equipment as live agents have meant that most task-oriented AI studies have been constrained to virtual simulation [34], [35], [36], where the simulated agent iteratively interacts with virtual surroundings. This approach has shown promise in other robotic domains where the environment is not as unpredictable and variable as a construction job site.

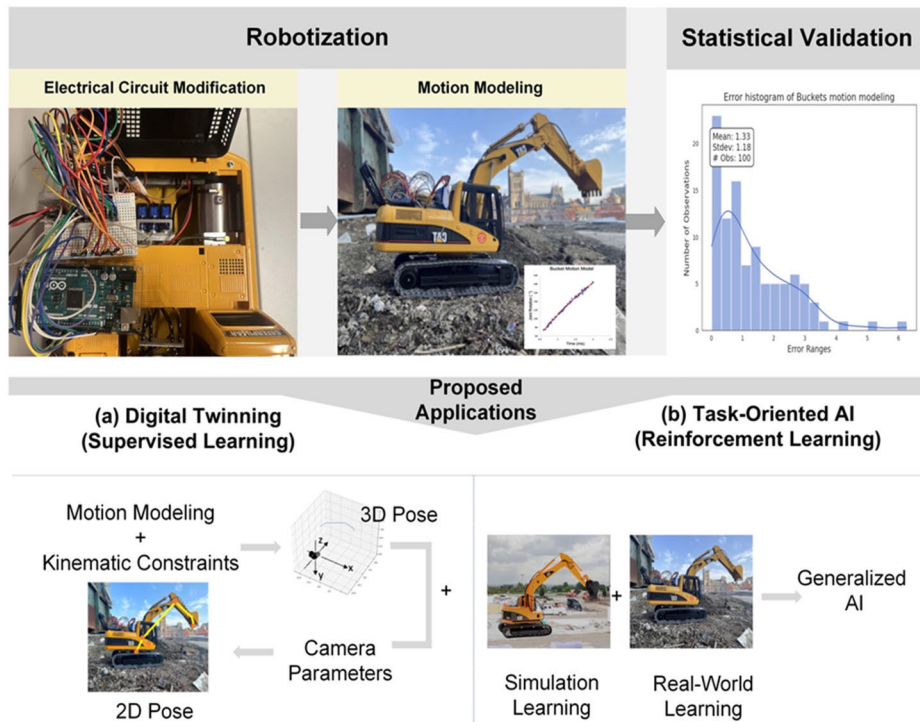


FIGURE 1. Overview of this study.

However, relying on simulation for training task-oriented AI for excavators entails serious limitations that greatly restrict the effectiveness of this approach.

The literature reveals that training in a preset virtual environment often fails to include real-world complexities [37]. Unlike controlled environments in industrial automation and robotics, construction site terrain variability, unevenness, and unpredictability make it difficult to accurately model different job site and excavator behaviour in a simulation environment [38], [39], [40]. Virtual simulation and training will continue to have a role in training task-oriented models as these tools accelerate the training process by harnessing computational power. However, training in a virtual environment with simplifications never suffices, and real-world interactions must be incorporated to capture the complex and dynamic states often found on construction sites [41]. Thus, a safe and practical live agent that can learn both in controlled (e.g., in a lab or a virtual environment) and wild (e.g., in real construction sites) environments is called for [37].

C. RESEARCH OBJECTIVES: ROBOTIZING AN RC-EXCAVATOR WITH VALIDATED MOTION ACCURACY

The research gaps identified in this study are (i) the absence of a feasible method to label excavators' 2D/3D poses and (ii) the lack of live training agents capable of self-training while interacting with real surroundings. To address these twin challenges, we introduce a novel approach using a miniature-scale RC-excavator, presenting a cost-effective

method to leverage robotization for both (i) automated data collection and labeling (Figure 1a) and (ii) interactive reinforcement learning (Figure 1b).

The chosen RC-excavator model, mirroring its full-scale counterpart in terms of appearance and kinematic structure, presents several advantages when compared to a full-scale excavator. These advantages include (i) The capability to accurately simulate diverse construction site scenarios within the limited confines of a controlled laboratory environment, (ii) The potential for round-the-clock simulation with minimal energy consumption and environmental footprint, and (iii) A complete elimination of operational hazards and safety risks inherent to large-scale equipment operations.

However, the operating system of the available RC-excavator had limitations which restricted its utility for our research objectives. A primary limitation was that its joystick-based motion control system lacked the capability to record angular displacements of individual excavator components. Moreover, achieving precise control using the handheld controller was challenging. Additionally, the original receiver-transmitter system was devoid of onboard computing hardware and software capabilities. As a result, executing a pre-programmed set of movements, let alone advanced functions like reinforcement learning, was not possible. To overcome these constraints, robotization was required, unlocking the use of the RC-excavator for both automated data collection and task-centric learning.

Robotization necessitates electrical hardware modifications, the establishment of a wireless connection between the

RC-excavator and a computing platform, and comprehensive system identification to comprehend and model the system's kinematic structure and motion patterns. Through robotization, the RC-excavator acquires the capabilities for precise motion control and onboard computing. This enhancement enables fully automated data collection and labeling while also allowing for reinforcement learning in a risk-free environment.

It's noteworthy that several prior studies have leveraged the inherent features of RC-excavators in varying contexts, confirming their potential as an effective tool for tasks akin to those of full-scale excavators [42], [43]. Such research has provided insights into the potential utility of miniature-scale construction equipment. Nevertheless, these past studies largely focused on tasks like autonomous continuous trajectory planning and Blended Shared Control (BSC) architecture [42], [43], [44]. These areas of focus do not holistically address the challenges tied to automated labeling for digital twinning nor the broader implications of excavator robotization. In contrast, our study aims to present a replicable solution in which the robotized RC-excavator's motion accuracy has been fully validated, thus paving the way for pose labeling and reinforcement learning in an in-lab setting.

II. THE ROBOTIZATION OF A MINIATURE-SCALE RADIO-CONTROLLED EXCAVATOR

Through a detailed examination of the electrical circuit and each component of the original RC-excavator, we identified electrical parts that needed engineering modifications. Required additions included (i) a receiver-transmitter system and (ii) wireless communication using a Bluetooth module. This section elaborates on our choice of model, explains the stock (as-purchased) setup and configuration, and details the end-to-end process of robotization.

A. MODEL SELECTION

We conducted an extensive market survey of commercially available RC-excavator models. Following the search and evaluation, we chose the Hydraulic CAT 320 Bruder Excavator manufactured by Magom HRC [45], as illustrated in Figure 2. Our decision was based on the following criteria:



FIGURE 2. Hydraulic CAT 320 bruder excavator, Magom HRC.

- **Mechanical structure:** The kinematic structure of the selected model mirrors that of a full-scale 40-ton excavator. Ensuring identical kinematic structures is a fundamental prerequisite for data collection and labeling, as it ensures that the model can accurately emulate the motions and tasks of a real-scale excavator.

- **Colour:** Yellow is a commonly chosen colour for construction equipment because it stands out against various backgrounds, thereby enhancing safety on construction sites. With this consideration, we selected the yellow-coloured Hydraulic CAT 320 Bruder Excavator.

- **Scale:** With a 1:16 scale, our selected model is compact for easy handling and modification, yet it's large enough to authentically replicate the functions of a full-size excavator. This scale offers an optimal balance between practicality and realism.

- **Power system:** Unlike full-size excavators, our chosen model is powered by electricity, running at a voltage of $U = 11.1$ V. Furthermore, the model's hydraulic consumers, such as the main arm, middle arm, and bucket, are powered by a single constant flow gear pump, driven by an electric motor. This compact yet effective power system allows the RC-excavator to function nearly identically to a real excavator across different soil conditions.

B. THE OPERATIONAL FEATURES OF THE SELECTED MODEL

The chosen model, the Hydraulic CAT 320 Bruder Excavator, is equipped with self-contained control and power systems with a sophisticated electric circuit. To advance with the robotization process, it was imperative to comprehend the model's essential components, their distinct features, and their inter-connections. We identified six pivotal systems, summarized as follows:

- **Controller and transmitter:** The controller and transmitter are the core command system of the original RC-excavator, allowing the manual control of the excavator. The transmitter generates a radio signal corresponding to the input from the control switches, which is then sent to the receiver on the RC-excavator.

- **Main cabin and receiver:** The receiver, housed in the main cabin, is the operational center of the original RC-excavator. It processes the signals transmitted by the transmitter, converting them into commands for various actions. Each of the receiver channels corresponds to a specific movement of a component. The receiver of our chosen model is a RadioLink R12DS [46], which comes with eight channels (see Table 1). Four channels are assigned to moving hydraulic-dependent components, which consist of the bucket, main arm, cabin, and pump. Three channels are assigned to hydraulic-independent movements, and one channel is free.

- **Electric Speed Controller (ESC) and an electric motor:** The ESC is an essential component connected to channels 4, 7, and 8. The forward/reverse brushed ESCs receive signals from the receiver through these channels, regulating the speed and direction of the corresponding motors. These electric motors are responsible for the rotation of the excavator's main cabin and the forward and backward movements of the left and right tracks [47].

- **Valve and pump (i.e., hydraulic system):** The hydraulic system is fundamental to the operation of the bucket and the

main arm. It involves valves, servos, and the hydraulic pump. The servos, responding to signals from the receiver, control the valves, which in turn manage the flow of hydraulic fluid. This regulated fluid flow drives the hydraulic pump, which powers the movements of the bucket and main arm.

- **Brushless Speed Controller (BSC) for the pump:** This speed controller, which operates directly with channel 6, regulates the pump speed and, thus, the hydraulic flow in response to commands from the receiver. The BSC drives the gear pump, which powers the hydraulic consumers, such as the main arm and the bucket. This component is crucial for any operation involving the hydraulic system.

- **Servos:** The servos are responsible for actuating the servo valves. They receive Pulse Width Modulation (PWM) signals from the receiver via channels 1 and 3, which correspond to the desired positions of the valves. Valves control the hydraulic oil flow rate into the corresponding excavator component, providing precise fluid flow control within the hydraulic system. The servo valves make this precise fluid flow control possible, allowing accurate velocity control of the main arm and the bucket [47]. To initiate the hydraulic system's fluid flow for these movements, channel 6 (i.e., the pump) must be activated. The valves are then controlled by the PWM signal from channels 1 and 3 to move the bucket and the main arm.

TABLE 1. 8-Channel configuration of the Radiolink(R12DS).

Channel	Component Connected	Neutral PWM	Operating PWM
1	Bucket	200	Open Bucket: 130 Closed Bucket: 250
2	Main Arm	180	Main Arm Down: 115 Main Arm Up: 250
3	Middle Arm	185	Middle Arm Down: 250 Middle Arm Up: 120
4	Rotation	185	Clockwise Rotation: 135 Counterclockwise: 240
5	Not in use	N/A	N/A
6	Motor	170	Activating Hydraulic System: 210
7	Left Track	180	Backward: 112.5 Forward: 250
8	Right Track	180	Backward: 112.5 Forward: 250

Note: PWM stands for Pulse Width Modulation, a technique used to control the power delivered to electronic devices, especially for driving motors in robots and RC vehicles. By modulating the pulse width in a PWM signal, the position, speed, or torque of servo motors can be dictated, enabling precise control of robotic components [48].

C. THE MODIFICATION OF THE ELECTRIC CIRCUIT

A primary goal of our robotization of the RC-excavator was to establish wireless communication with a computer such that commands can be managed and stored in the computer. To this end, we engineered the stock excavator model, replacing the original receiver-transmitter system, adding additional payloads, including a Bluetooth module,

and modifying the base electric circuit. The detailed process is as follows (see Figure 3):

- **Adding a programmable circuit board – Arduino:** We replaced the built-in receiver of the model (R12DS) with an Arduino Mega 2560 microcontroller [49]. This modification was crucial as it allowed for controlling the excavator through commands issued for the serial interface of a computer [48]. Consequently, each joint's angular displacement can be dictated directly from the computer, eliminating manual intervention and making motion fully programmable.

- **Adding a wireless connection module – Bluetooth:** We integrated a Bluetooth module HC-05 [50] into the electrical circuit to enable wireless connectivity. Operating with a baud rate of 9,600 (a measure of data transmission speed), this Bluetooth module facilitates serial communication between the Arduino microcontroller and the laptop. Bluetooth offers superior power efficiency compared to Wi-Fi modules, reducing power consumption. This ensures optimal battery utilization without compromising the excavator's performance. In addition, Bluetooth enables a direct device-to-device connection, eliminating the need for additional network infrastructure. This option mitigates potential disruptions from unstable Wi-Fi connections, providing enhanced reliability and stability. The ability to connect an external computer directly with the programmable circuit board (i.e., Arduino Mega 2560) enables the system to conduct pre-programmed and real-time excavator operations.

- **Adding additional resistors:** To ensure compatibility with the HC-05 Bluetooth module, we added resistors of 1,000 Ω , 330 Ω , and 220 Ω to the input signal of the Bluetooth module. This setup created a voltage divider converting the 5V output of the Arduino TX pin to the necessary input voltage of the HC-05, which is less than 3.3V.

These modifications provided the necessary platform for precise remote control of the RC-excavator. It is worth noting that the power and hydraulic systems of the original RC-excavator remained intact, maintaining the integrity of its mechanical operations.

D. MOTION MODELING OF THE ROBOTIZED RC-EXCAVATOR

Understanding the relationship between the commands (e.g., signal duration) and the joint movements (e.g., angular shift) is crucial for accurately predicting and controlling the robotized RC-excavator. This requires constructing a mathematical model representing how each joint responds to a given PWM signal value and duration. To this end, we conducted the following procedure and attained our best-fit model per each joint:

- **Data collection:** Initially, we observed the full range of angles that each excavator component could undertake. The bucket had a full range of 154°, the main arm 60°, and the cabin rotation spanned a full circle of 360°. We collected 100 samples for each component evenly distributed within its movement range. This approach ensured that crit-

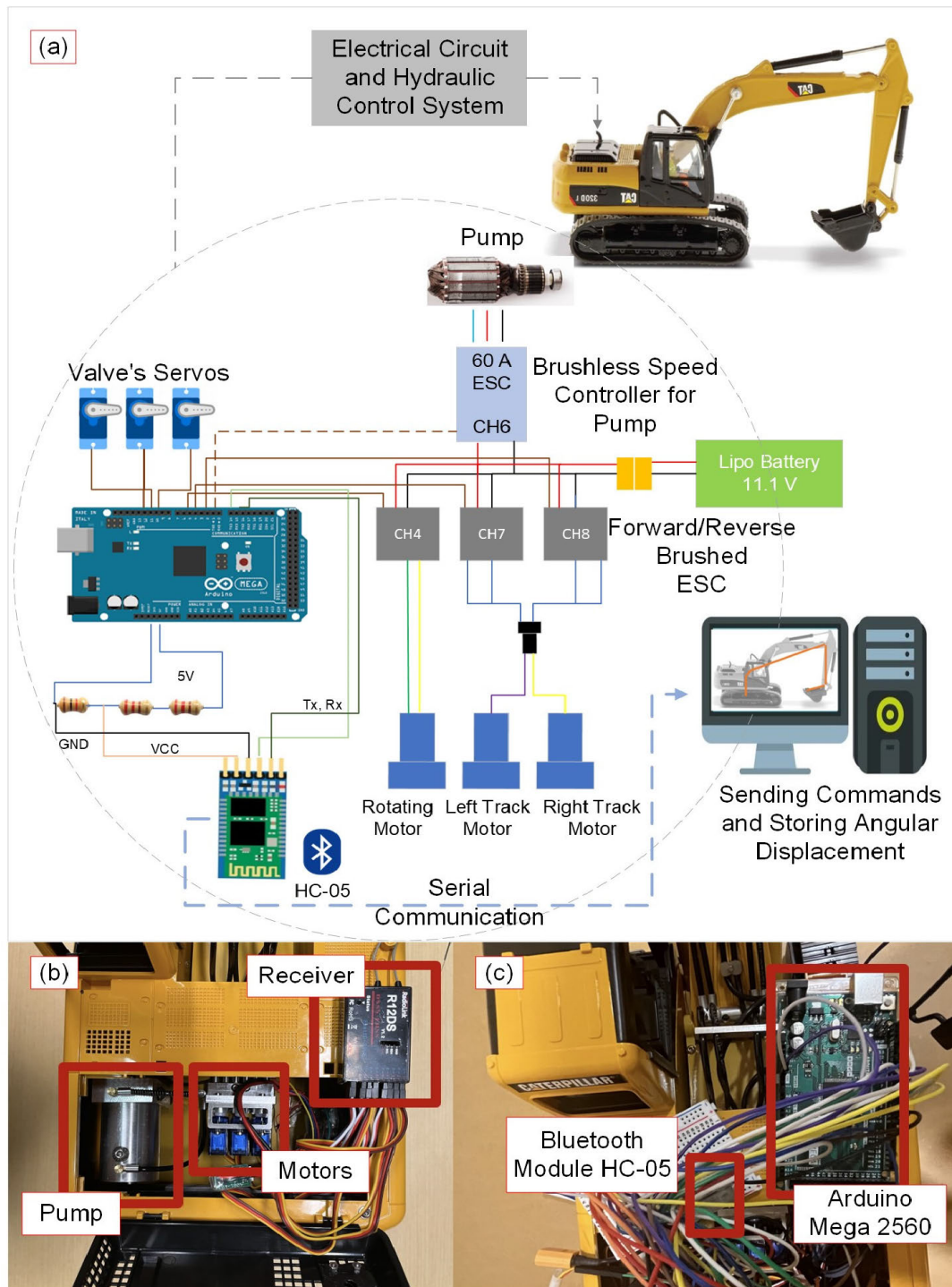


FIGURE 3. Modifications: (a) Schema, (b) Original circuit, and (c) Modified circuit.

ical positions, transitions, and movement patterns were all represented. Following data collection, we screened the samples to exclude outliers and maintain data quality, retaining only those that reflected typical operation. Given the straightforward mechanics and limited variability of the 1:16 RC excavator, this dataset provides sufficient resolution for modeling component behavior accurately. These observations

were measured manually using a digital protractor with $\pm 0.3^\circ$ accuracy. Each record contains input values (i.e., signal duration) and the corresponding angular displacements of each excavator joint. Table 1 shows the operating and neutral PWM signals for different components of the excavator. “Neutral PWM” refers to a baseline signal level at which the excavator components remain inactive - essentially keeping

them at rest. For operations involving components such as the bucket and arms that require hydraulic activation, the pump responsible for hydraulic system must be activated by receiving a minimum PWM signal to activate the hydraulic system. To keep the hydraulic system consistently active throughout the experiments, we maintained a constant PWM value of 210. By keeping the motor at a constant 210 PWM, we ensured that the hydraulic system was always ready, thus avoiding delays or interruptions in the activation of the bucket, and the main arm.

- **Method – Regression Modeling:** We selected regression modelling to mathematically model the relationship between the duration of transmitting signals and corresponding motions [51]. Our empirical experiments started with a first-order linear regression model; however, an examination of plots showed a U-shaped pattern in residuals, suggesting that other models could improve the fit to the data. As a result, higher-order polynomial models were added for consideration. A comprehensive analysis of polynomial regression models was done by comparing statistical metrics such as R^2 , *Standard Deviation of Error (SDE)* and *Mean Absolute Error (MAE)* for each excavator component [51]. Our analysis of different-order regression models led us to conclude that a second-order polynomial best predicted the movements of the joints, as summarized in Table 2. Although, in some cases, the third-order polynomial yielded similar results or showed a marginal improvement in *MAE*, we opted for the quadratic regression models, as not only is interpretation easier with simpler regression models, but the risk of over-fitting is also mitigated. Our empirical tests showed that the marginal gains in accuracy from higher-order models did not justify the added complexity and risk of overfitting. Statistical validation of these results is detailed in Section III.

TABLE 2. Regression modeling results: First-, Second-, and Third-order polynomial models.

Excavator Component	Motion Model	R^2	MAE
	$\theta = 79.18 t + 29.14$	0.99	1.83
Bucket	$\theta = -11.81 t^2 + 109.52 t + 11.21$	0.99	1.33
	$\theta = -0.01 t^3 - 11.75 t^2 + 109.45 t + 11.24$	0.99	1.33
	$\theta = -40.77 t + 75.85$	0.98	1.56
Main Arm	$\theta = 8.101 t^2 - 58.232 t + 84.198$	0.98	1.21
	$\theta = -5.582 t^3 + 26.312 t^2 - 76.477 t + 89.69$	0.98	1.23
	$\theta = 72.55 t + 16.93$	0.99	5.47
Cabin	$\theta = 1.76 t^2 + 63.90 t + 23.48$	0.99	4.92
	$\theta = 0.39 t^3 - 1.12 t^2 + 69.49 t + 21.25$	0.99	4.82

This setup allows us to control the RC-excavator effectively (Figure 3). For example, the excavator could perform a pre-programmed task of 20° counterclockwise motion for

the main arm, followed by a 10° clockwise rotation for the bucket. Any series of motions can be readily simulated, including simultaneous movements of multiple components.

E. NEW COMMAND SYSTEM

The regression-based movement models from Table 2 became the basis for a new command system, harnessing this knowledge of the correlation between the PWM signal duration and the joint rotation of each component to generate precise, repeatable motions (see Figure 4). This command system allows the programmer to specify a particular angular displacement of an excavator component. The system then calculates the requisite PWM signal duration to realize this displacement and relays this command to the excavator. As a result, we can specify an exact degree of rotation for each component, greatly improving our control and precision.

This enhancement to the original command system enables the miniature-scale robotized RC-excavator to execute more intricate and accurate tasks in a highly repeatable manner. For example, it can now perform all motions needed for a complete digging cycle, trenching, and more, all with a few lines of commands in a script.

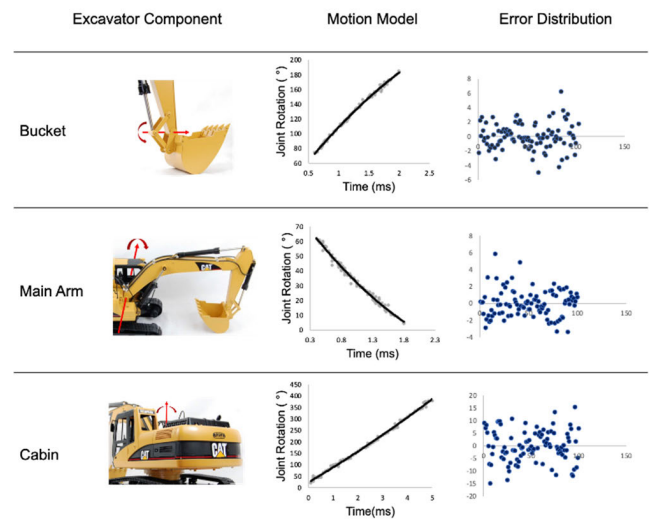


FIGURE 4. Fitting quadratic regression line to motion data of the bucket, main arm, and cabin.

III. STATISTICAL VALIDATION OF THE MOVEMENT ACCURACY

Receiving a detailed command (e.g., a 20° clockwise movement of the bucket) and executing the precise corresponding action are core capabilities of the robotized RC-excavator. Without these features, automatic and simultaneous data collection and labeling, as well as interactive reinforcement learning, would be fatally compromised. Therefore, this study aimed to meticulously validate the movement accuracy of the robotized RC-excavator. We also assessed its repeatability, gauging the consistency of output results generated by our

new command system. To validate the repeatability of the RC-excavator, measurements were taken of repeated actions where the same input was administered under identical conditions. The degree of variation in these measurements was then analyzed using multiple metrics.

A. VALIDATION METRICS

To quantitatively evaluate the accuracy of the movement of the robotized RC-excavator we employed the following metrics: (i) *Normalized Range-Adjusted Accuracy (NRAA)*, (ii) *Mean Absolute Error (MAE)*, (iii) *Standard Deviation of Errors (SDE)* and (iv) *Coefficient of Determination (R^2)* which are represented in (1) to (4) respectively.

- *Normalized Range-Adjusted Accuracy (NRAA)*: This metric measures accuracy by considering not just the deviation from actual values but also in relation to the maximum movement range of each component. Computing the *NRAA* involves (i) calculating the absolute error between predicted and actual angular displacements for each observation (ii) normalizing this error by dividing it by the component's maximum possible movement range, giving the error as a proportion of this total range; (iii) subtracting this proportion from 1 to provide an accuracy measure for each prediction; and lastly (iv) computing the average of these individual accuracies to yield an overall accuracy measure for each component of the robotized RC-excavator. By doing so, *NRAA* offers a context-aware assessment of accuracy, indicating how closely the predicted displacements align with actual displacements, given the maximum potential movement.

$$NRAA (\%) = \frac{\sum_{i=1}^n 1 - \frac{|\theta_i - \theta_p|}{\text{MaximumRangeofMotion}}}{n} \times 100, \quad (1)$$

where θ_i refers to the actual value, θ_p refers to the predicted value, and n shows the total number of observations.

- *Mean Absolute Error (MAE)*: This metric calculates the average of the absolute differences between predicted and actual values, offering a direct estimate of the system's accuracy in responding to input commands. A lower *MAE* value indicates a higher degree of accuracy [51].

$$MAE = \frac{|\theta_i - \theta_p|}{n}, \quad (2)$$

where θ_i refers to the actual value, θ_p refers to the predicted value, and n shows the total number of observations.

- *Standard Deviation of Errors (SD)*: This metric quantifies the dispersion of error values around their mean. A lower *SDE* suggests that errors are tightly clustered, indicating a precise and consistent estimation model. Conversely, a higher value indicates varied prediction accuracy, suggesting potential inconsistencies. Evaluating this dispersion serves as an insight into the model's robustness across diverse input settings. It also acts as a diagnostic tool, pointing to underlying biases or overlooked non-linear relationships, which could

guide future model refinement [51].

$$\sigma = \sqrt{\frac{\sum (\theta_i - \bar{\theta})^2}{n - 1}}, \quad (3)$$

where $\bar{\theta}$ refers to the mean value of all observations.

- *The coefficient of Determination (R^2)*: This metric signifies the proportion of the variance in angular displacement, which is represented by the duration of signal transmission in the regression model. A higher R^2 value denotes a closer fit of the model to the observed data, suggesting that the model can more accurately predict the output based on its input variables [51].

$$R^2 = 1 - \frac{\sum (\theta_i - \theta_p)^2}{\sum (\theta_i - \bar{\theta})^2}, \quad (4)$$

where θ_i refers to the actual value, θ_p refers to the predicted value, and n shows the total number of observations. $\bar{\theta}$ refers to the mean value of all observations.

B. VALIDATION RESULTS

Table 3 summarizes the evaluation result of each component of the robotized RC-excavator.

TABLE 3. Summary of validation metrics.

Excavator Component	NRAA	MAE	SDE	R^2
Bucket	99.14%	1.33°	1.79°	0.99
Main Arm	97.97%	1.21°	1.61°	0.98
Cabin	98.63%	4.92°	6.18°	0.99

- *NRAA* serves as an assessment metric, quantifying the accuracy of our motion modelling for all the robotized RC-excavator components relative to their maximum movement range. The respective accuracies are reported as 99.14% for the bucket, 97.97% for the main arm, and 98.63% for the cabin. Such elevated accuracies, especially in comparison to the accuracy of conventional labeling methods, not only validate our approach but emphasize its reliability.

- *MAE* is the other chosen quantitative metric which provides insight into the accuracy of our regression modelling. An *MAE* of 1.33° for the bucket showcases a low estimation error. The main arm motion modelling is similarly precise with an *MAE* of 1.22°. However, while it has a 360° rotation range, the cabin's rotation modelling exhibits a slightly higher *MAE* of 4.92°. Despite this, given the complexity of the cabin's range, this value indicates that the accuracy of cabin motion will support the use of the RC-excavator as an effective medium for pose labeling and reinforcement learning live agent.

- R^2 affirms the robustness of our motion modelling. R^2 values of 0.99 for the bucket, 0.98 for the main arm, and 0.99 for the cabin underscore the strength of our model's fit to the data. Such high R^2 values indicate the model was able to

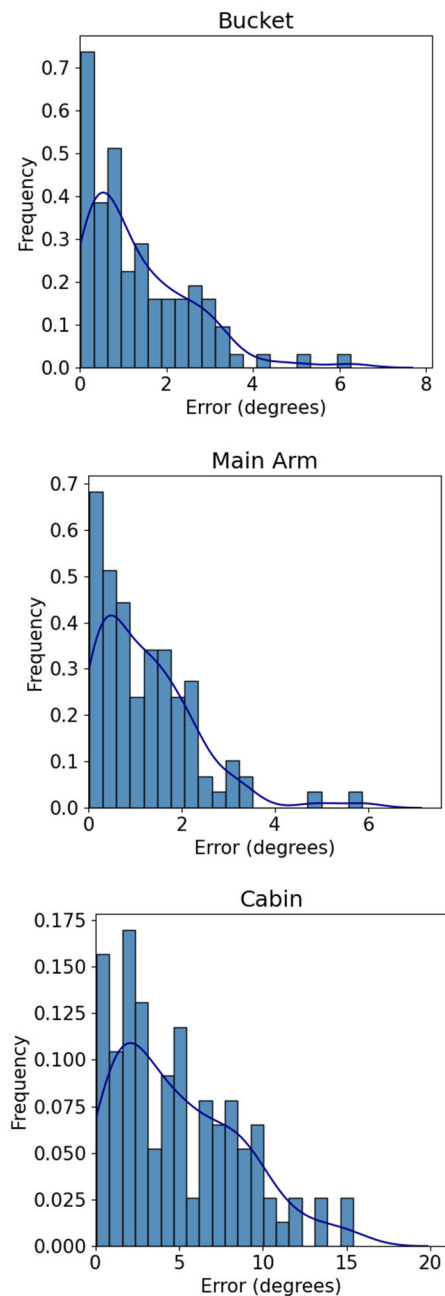


FIGURE 5. Error histogram for the bucket, the main arm, and the cabin.

accurately capture the inherent variability of the motion data (Table 3).

• Lastly, the *SDE* provides a measure of consistency in our predictions. With values of 1.79° for the bucket, 1.61° for the main arm, and 6.18° for the cabin, these metrics suggest a tightly clustered error distribution around the mean. The relatively higher *SDE* for the cabin is likely attributable to its extensive 360-degree range of movement while bearing the weight of all other components. In practical terms, this implies that our model performs consistently across different scenarios, reinforcing the reliability of the angular predictions for all excavator components (Figure 5).

Our validation analysis indicated a high degree of accuracy in our robotized RC-excavator system, as illustrated in the histograms of the absolute value of errors for each excavator component (see Figure 5 and Table 4). For the bucket, 71% of attempts had errors less than 1.82° . However, higher errors were observed in 4% of attempts, ranging from 3.62° to 6.32° . For the main arm observation, 71% of attempts exhibited errors of less than 1.59° , indicating a very accurate estimation with our model. Nevertheless, higher errors were observed in a subset of attempts, reaching a maximum of 6.33° . Similarly, the rotation of the main body with a full range of 360° showed accurate performance, with 65% of attempts achieving errors less than 5.67° . However, a few attempts exhibited errors from 11.27° to a maximum of 16.87° . These results offer essential insights regarding the precision of the labels generated by our command system. While the model consistently performs standard operations with a high degree of accuracy, the presence of outliers in a few instances underscores a potential need for recalibration when executing more intricate maneuvers.

TABLE 4. Distribution of errors.

Bucket		Main Arm		Cabin	
Error Range	Frequency	Error Range	Frequency	Error Range	Frequency
$< 0.92^\circ$	51	$< 0.80^\circ$	43	$< 2.87^\circ$	37
$0.92^\circ - 1.82^\circ$	20	$0.80^\circ - 1.59^\circ$	27	$2.87^\circ - 5.67^\circ$	28
$1.82^\circ - 2.72^\circ$	16	$1.59^\circ - 2.38^\circ$	20	$5.67^\circ - 8.47^\circ$	17
$2.72^\circ - 3.62^\circ$	9	$2.38^\circ - 3.17^\circ$	6	$8.47^\circ - 11.27^\circ$	11
$3.62^\circ - 6.32^\circ$	4	$3.17^\circ - 6.33^\circ$	4	$11.27^\circ - 16.87^\circ$	7

To gauge the repeatability of our robotic system, we computed the standard deviations of output joint angles for repeated identical inputs over a series of trials (see Table 5). These inputs were consistently applied for a duration of 1,200 milliseconds. A smaller standard deviation implies that the robot's performance is highly consistent and exhibits minimal variability for repeated inputs. For example, the bucket showcased a high degree of repeatability with a standard deviation of just 1.18° , underscoring its predictable behavior under consistent conditions. The main arm further exemplified this consistency, presenting an even tighter deviation of 0.60° for the same input duration. Meanwhile, the cabin rotation displayed a slightly larger standard deviation of 6.14° . However, given the cabin's intricate and broad range of motion, this result was considered acceptable.

In conclusion, our analysis revealed that the main discrepancies in motion estimation stem from the quality and

calibration of the servo motors. The vendor-installed servos might have deviated from their initial calibration over time due to potential quality issues, resulting in mismatches between commanded and actual movements. Additionally, it's worth noting that our methodology predominantly relied on forward kinematic modelling of the excavator. The omission of dynamic system modelling could have influenced the kinematic performance, further contributing to the observed errors.

TABLE 5. Standard deviation of joint angles under constant input signals.

Excavator Component	Time (milliseconds)	PWM Signal	Joint Angle's Standard Deviation
Bucket	1200	130	1.18°
Main Arm	1200	115	0.60°
Cabin	1200	135	6.14°

While the observed discrepancies have a minimal impact on the system's efficacy for data collection, labeling, and reinforcement learning, there are avenues for further optimization. Considerations include A) adopting a closed-loop system with joint encoders to counteract servo motor-induced errors and B) performing pre-installation calibration of the servos. Additionally, leveraging reinforcement learning for motion planning could inherently adjust for unforeseen dynamics. Future studies will aim to refine these aspects to enhance accuracy further. In addition, the middle arm of the RC-excavator was found to be mechanically malfunctioning from the initial unboxing stage of the equipment. The functionality of the middle arm is independent and does not interact with the other components under study. Despite this limitation, the functional components (bucket, main arm, cabin) provided sufficient data to validate our approach effectively.

IV. DISCUSSION: AUTOMATED 2D AND 3D POSE LABELING AND REAL REINFORCEMENT LEARNING

In this discussion, we first present a framework for automating the process of 2D and 3D pose labeling, situated within our envisioned RC-excavator based data collection environment. We also delve into the potential benefits of incorporating new modalities into regression learning methodologies, discussing the viability of using RC-excavator images as substitutes for real-scale images in DNN training. Secondly, we investigate the prospects of employing the RC-excavator robot as an agent for the application of hybrid reinforcement learning (hybrid-RL). A hybrid-RL approach leverages high-quality offline data to bridge the gap between virtual learning and real-world excavator operation.

A. AUTOMATED IMAGERY DATA COLLECTION AND 2D/3D POSE LABELING

The robotized RC-excavator is an ideal tool for in-lab data collection and automated labeling. The resulting dataset consisting of images with 2D and 3D pose labels would directly address the data scarcity problem that limits the training of excavator-specific AI and digital twinning. Figure 6 illustrates a visual proof-of-concept, displaying both 2D and 3D visualizations of the excavator pose, i.e., skeleton, based on the Denavit–Hartenberg (DH) convention [52].

The hardware setup for automated imagery collection and pose labeling encompasses a set of cameras, a robotized RC-excavator, and a dedicated space for the robotized RC-excavator to carry out varied tasks. Given that this data is intended to serve as ground truth labels, ensuring precise camera parameter settings is fundamental. It's crucial to calibrate each camera's intrinsic and extrinsic parameters to ensure the images maintain the correct scale and orientation. Intrinsic parameters denote a camera's internal characteristics, such as focal length and optical center. A camera's extrinsic parameters relate to its 3D spatial position and orientation. In addition, temporal synchronization is vital when deploying multiple cameras. Lastly, lighting conditions need to be adjusted to replicate those found in actual outdoor construction sites as closely as possible. The dataset derived from this meticulously designed hardware setup featuring a robotized RC-excavator can bridge the existing gap in excavator-specific image datasets. Integrating additional modalities into the data collection chamber presents additional exciting avenues to explore in excavator digital twinning. For example, augmenting our setup with depth and infrared sensors alongside the existing RGB images could significantly enrich the data we capture, offering more comprehensive input channels for DNN training. Utilizing RGB-D images compensates for any absence of ground-truth depth data, thereby elevating the quality of our training dataset and, by extension, the efficacy of our deep-learning models. Moreover, combining RGB with infrared (RGB+IR) data can pave the way for specialized deep-learning methods tailored for night vision monitoring. Crucially, the proposed data collection method would facilitate such enhancements without the need for substantial modifications, requiring only the calibration and synchronization of the newly added sensors.

The process of mapping a 3D world onto a 2D image plane through a monocular lens does not capture scene-scale data. A camera-captured image cannot tell if the objects being captured are at a scale of meters or millimeters. Therefore, images of miniature-scale RC construction equipment, when photographed properly with respect to the camera's distance, can be as real as images of their full-scale counterparts. Within our data collection chamber, we can obtain a diverse set of training images of an excavator under a myriad of imaging conditions that include variations in scene scale, viewpoint, and illumination. Complementing this collection

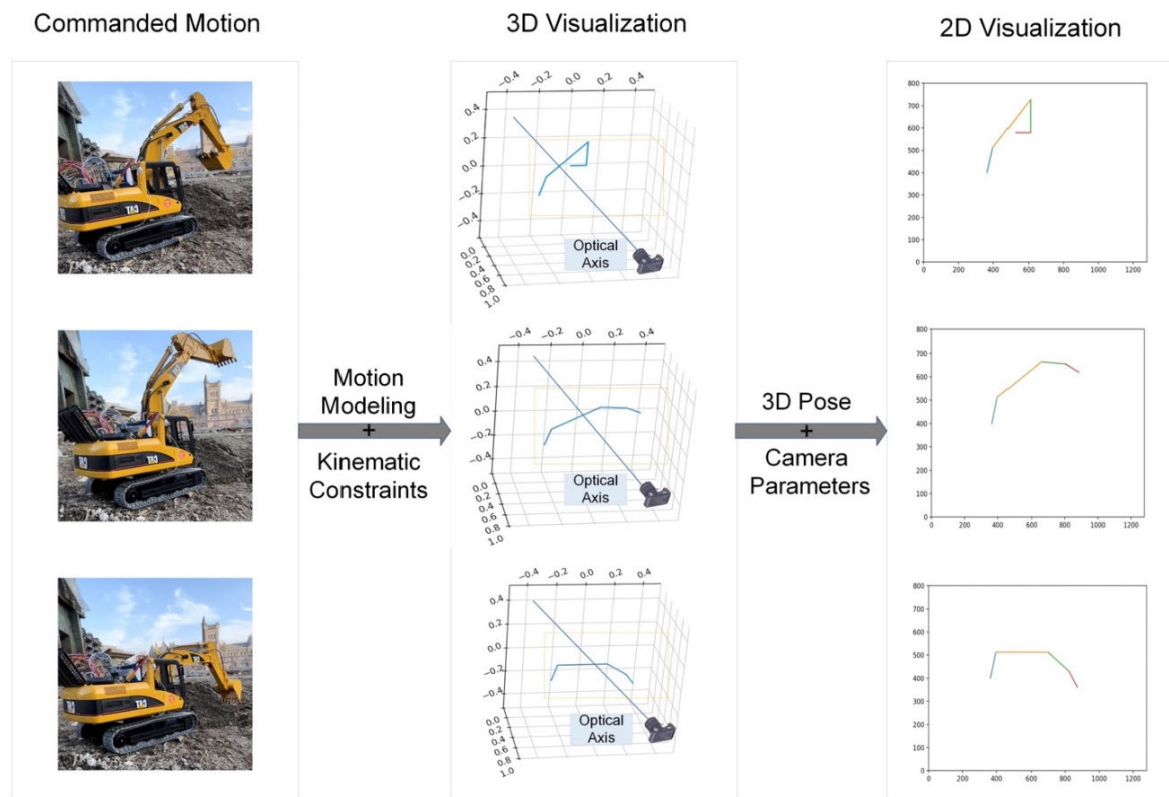


FIGURE 6. Proof of concept, 2D and 3D visualization of excavator skeleton.

with an equivalent volume of depth and thermal images will further enrich our dataset, providing additional layers of input for DNN training.

While there is recognized value in using synthetic images for training DNNs, we anticipate that using the miniature-scale RC-equipment images that feature more realistic images than those generated synthetically will yield significantly better results. Although the current study does not explore this in-depth, the principles of scale invariance, coupled with the known advantages of realistic images over synthetic ones, reinforce our confidence in this approach. The next phase of our research will encompass multiple rounds of data collection and labeling, DNN training for tasks like 2D/3D object detection and 2D/3D pose estimation, hyper-parameter optimization, and finally, field testing and demonstration.

B. REAL REINFORCEMENT LEARNING

Previous research in excavator-specific AI has predominantly relied on simulated environments to develop task-oriented AI algorithms. These algorithms typically exhibit a protracted learning curve due to the inadequacies of synthetic images as training datasets. Attempts to address the limitations of training on digitally simulated images have exploited advances in computational capacity to better mimic real-world operations

within responsive virtual settings. However, the efficacy of these models when transitioned to real-world scenarios has shown to be underwhelming, with the multifaceted realities of the physical world not accurately replicated in these simulations [53]. Our strategy of coupling the learning process of AI algorithms with data generated by our robotized excavator system promises to significantly reduce the time required for these algorithms to attain peak performance.

After the preliminary development and testing of our task-oriented AI in a simulation, our next step will be its training and evaluation in the RC-excavator environment. By training the AI through a real agent immersed in various scenarios where it encounters diverse soil types, terrains, and other conditions, this approach stands to profoundly enhance the degree of comprehension and functionality achieved as compared to training through digital simulation.

With the understanding that simulation-based pre-training can contribute to the reinforcement learning process, the authors call for attention to hybrid reinforcement learning [54]. Hybrid-RL integrates offline high-quality datasets with online real-world experiences to overcome the hurdles seen in purely offline task-oriented settings, potentially leading to the development of efficient and resilient algorithms. An offline dataset in this context would consist of a collection of labelled data from our robotized RC-excavator,

which details the full-skeleton poses while it performs various tasks in different soil types and terrain slopes. This offline dataset is the RL agent's reference or initial knowledge base. However, relying solely on the offline dataset would limit the RL agent's performance, weakening its ability to adapt to new or unseen situations. This is where online RL comes into play. As our robot interacts with various environments, it can learn from these online experiences, adjusting its policy to optimize for better performance [54].

The RC-excavator system detailed here has the potential to realize a hybrid approach which deploys real reinforcement learning to rapidly develop excavator-specific AI which can adapt and learn from its surroundings, facilitating efficient and practical real-world applications. In our continuing research, we will investigate the potential of the robotized RC-excavator system to support advanced hybrid reinforcement learning.

C. METHOD GENERALIZABILITY

An essential aspect of our work is its generalizability to other RC-equipment that features articulated body parts. Although our focus has been on excavators, the core approach involving 3D pose labeling using kinematic modeling of RC-equipment, camera projection, and other processes can be transferred to different types of machinery. The proposed method is adaptable to a wide range of RC equipment, making it suitable for data generation and reinforcement learning in an in-lab setting. Although our study centers on the Magom 1:16 excavator, this method is broadly applicable due to the common architecture of RC systems, which typically include a transmitter and receiver. This shared system architecture allows for the electrical modifications we proposed to be adapted to other RC equipment with minimal changes, regardless of the vendor. Specifically, the consistency in receiver-transmitter systems and PWM signal processing across different models facilitates a straightforward transfer of our developed solutions to other RC platforms. The current trend in construction robotics involves retrofitting existing equipment with autonomous kits, such as the Exosystem by Built Robotics [55], enabling these machines to operate autonomously. This trend suggests that even future autonomous machines will likely resemble today's equipment in structure and appearance. Consequently, the demand for datasets specific to such equipment is expected to remain, highlighting the relevance of our approach.

In this study, we have laid the groundwork by developing a system that can be adapted to different construction machines. We aim to extend this research by using the generated dataset to train visual AI models, followed by testing their performance on real construction sites. In addition, we plan to further develop the reinforcement learning (RL) models and evaluate their effectiveness in real-world construction environments. This next step will allow us to better understand how the system represents real-world conditions and adapt it accordingly.

D. CHALLENGES IN ROBOTIZATION OF RC-EQUIPMENT

To extend the approach of robotizing miniature-scale RC equipment to other models or other machinery, given the common setup of receiver and transmitter, it is essential to acknowledge the technical challenges inherent to this process and explain the solutions we opted and implemented.

- **Serial Communication:** Correctly setting the COM port and synchronizing baud rates on both the computer and Arduino microcontroller is critical to ensure a reliable serial communication; without this, stable signal transmitting would not be possible.
- **Power Management:** It is essential to fully charge batteries before experiments. To maintain a stable power supply, we used fully charged batteries within the first 15 minutes for every experiment, while monitoring their status with a voltmeter; failing to do so could result in inconsistent performance.
- **Bluetooth Communication:** For the purpose of wireless communication with the equipment, a Bluetooth module was opted over Wi-Fi in order to avoid internet network instability. Bluetooth signal interference could result in unreliable communication; thus, the experiments were conducted in a lab environment with no other Bluetooth usage. Latency was a concern, but it was managed by allowing sufficient time for the machine to process commands in the experiments. Also, the HC-05 module was positioned within its optimal range (typically up to 8 meters) from the computer and a clear line of sight was maintained to ensure effective wireless communication.

These challenges have been systematically addressed, Resulting in enhanced robustness and reliability of our robotization approach. The solutions implemented underscore the versatility and adaptability of our method, making it applicable not only to the current model but also to a broad spectrum of RC equipment.

V. CONCLUSION

In this research, the authors have presented a detailed process for transforming an RC-excavator into a robotized live agent with validated motion modeling. The high accuracy of the RC-excavator's motion, as demonstrated by MAEs of 1.33° for the bucket, 1.22° for the main arm and 4.92° for the cabin, highlights the robotized RC excavator's potential as (i) a medium that can generate construction equipment benchmark datasets, and (ii) a live agent that enables reinforcement learning incorporating both real environments and hybrid approaches that combine offline and online learning. It should be noted that the proposed robotization can also be applied to other heavy construction machinery with known kinematic constraints.

Upcoming research will address two sources of error that this study did not fully address: (i) mechanical limitations due to poor quality and calibration in the servos and (ii) systematic limitations due to a lack of dynamic motion modeling. With these mechanical and systemic limitations addressed, we will embark on two research agendas—(i) DNN training

with miniature-scale data and (ii) live reinforcement learning leveraging the capabilities of the robotized RC-excavator.

REFERENCES

- [1] S. N. Naghshbandi, L. Varga, and Y. Hu, "Technologies for safe and resilient earthmoving operations: A systematic literature review," *Autom. Construction*, vol. 125, May 2021, Art. no. 103632, doi: [10.1016/j.autcon.2021.103632](https://doi.org/10.1016/j.autcon.2021.103632).
- [2] L. Bobadilla, A. Mostafavi, T. Carment, and S. Bista, "Predictive assessment and proactive monitoring of struck-by safety hazards in construction sites: An information space approach," *Comput. Civil Building Eng.*, vol. 2014, pp. 989–996, Jun. 2014.
- [3] B. Mahmood, S. Han, and J. Seo, "Implementation experiments on convolutional neural network training using synthetic images for 3D pose estimation of an excavator on real images," *Autom. Construction*, vol. 133, Jan. 2022, Art. no. 103996, doi: [10.1016/j.autcon.2021.103996](https://doi.org/10.1016/j.autcon.2021.103996).
- [4] L. Wen, D. Kim, M. Liu, and S. Lee, "3D excavator pose estimation using projection-based pose optimization for contact-driven hazard monitoring," *J. Comput. Civil Eng.*, vol. 37, no. 1, Jan. 2023, Art. no. 04022048, doi: [10.1061/\(ASCE\)CP.1943-5487.0001060](https://doi.org/10.1061/(ASCE)CP.1943-5487.0001060).
- [5] P. Egli and M. Hutter, "Towards RL-based hydraulic excavator automation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2020, pp. 2692–2697, doi: [10.1109/IROS45743.2020.9341598](https://doi.org/10.1109/IROS45743.2020.9341598).
- [6] I. Jebbor, Z. Benmamoun, and H. Hachmi, "Revolutionizing cleaner production: The role of artificial intelligence in enhancing sustainability across industries," *J. Infrastructure, Policy Develop.*, vol. 8, no. 10, p. 7455, Sep. 2024.
- [7] H. Omrany, K. M. Al-Obaidi, A. Husain, and A. Ghaffarianhoseini, "Digital twins in the construction industry: A comprehensive review of current implementations, enabling technologies, and future directions," *Sustainability*, vol. 15, no. 14, p. 10908, Jul. 2023, doi: [10.3390/su151410908](https://doi.org/10.3390/su151410908).
- [8] A. Assadzadeh, M. Arashpour, H. Li, R. Hosseini, F. Elghaish, and S. Baduge, "Excavator 3D pose estimation using deep learning and hybrid datasets," *Adv. Eng. Informat.*, vol. 55, Jan. 2023, Art. no. 101875, doi: [10.1016/j.aei.2023.101875](https://doi.org/10.1016/j.aei.2023.101875).
- [9] A. Assadzadeh, M. Arashpour, I. Brilakis, T. Ngo, and E. Konstantinou, "Vision-based excavator pose estimation using synthetically generated datasets with domain randomization," *Autom. Construct.*, vol. 134, Feb. 2022, Art. no. 104089, doi: [10.1016/j.autcon.2021.104089](https://doi.org/10.1016/j.autcon.2021.104089).
- [10] D. C. Marcu and C. Grava, "The importance of data quality in training a deep convolutional neural network," in *Proc. 17th Int. Conf. Eng. Modern Electr. Syst. (EMES)*, Jun. 2023, pp. 1–4, doi: [10.1109/EMES58375.2023.10171785](https://doi.org/10.1109/EMES58375.2023.10171785).
- [11] J. Kim, D. Kim, S. Lee, and S. Chi, "Hybrid DNN training using both synthetic and real construction images to overcome training data shortage," *Autom. Construction*, vol. 149, May 2023, Art. no. 104771, doi: [10.1016/j.autcon.2023.104771](https://doi.org/10.1016/j.autcon.2023.104771).
- [12] M. M. Soltani, Z. Zhu, and A. Hammad, "Skeleton estimation of excavator by detecting its parts," *Autom. Construct.*, vol. 82, pp. 1–15, Oct. 2017, doi: [10.1016/j.autcon.2017.06.023](https://doi.org/10.1016/j.autcon.2017.06.023).
- [13] W. T. Calderon, D. Roberts, and M. Golparvar-Fard, "Synthesizing pose sequences from 3D assets for vision-based activity analysis," *J. Comput. Civil Eng.*, vol. 35, no. 1, Jan. 2021, Art. no. 04020052.
- [14] F. Vahdatikhaki and A. Hammad, "Dynamic equipment workspace generation for improving earthwork safety using real-time location system," *Adv. Eng. Informat.*, vol. 29, no. 3, pp. 459–471, Aug. 2015, doi: [10.1016/j.aei.2015.03.002](https://doi.org/10.1016/j.aei.2015.03.002).
- [15] Y. Guo, H. Cui, and S. Li, "Excavator joint node-based pose estimation using lightweight fully convolutional network," *Autom. Construction*, vol. 141, Sep. 2022, Art. no. 104435, doi: [10.1016/j.autcon.2022.104435](https://doi.org/10.1016/j.autcon.2022.104435).
- [16] C. Lugaresi, J. Tang, H. Nash, C. McClanahan, E. Uboweja, M. Hays, F. Zhang, C.-L. Chang, M. Guang Yong, J. Lee, W.-T. Chang, W. Hua, M. Georg, and M. Grundmann, "MediaPipe: A framework for building perception pipelines," 2019, *arXiv:1906.08172*.
- [17] T. Jiang, P. Lu, L. Zhang, N. Ma, R. Han, C. Lyu, Y. Li, and K. Chen, "RTMPose: Real-time multi-person pose estimation based on MMPose," 2023, *arXiv:2303.07399*.
- [18] A. V. S. Abhishek and S. Kotni, "Detectron2 object detection & manipulating images using cartoonization," *Int. J. Eng. Res. Technol. (IJERT)*, vol. 10, no. 8, pp. 1–5, Aug. 2021. [Online]. Available: www.ijert.org
- [19] D. Pavlo, C. Feichtenhofer, D. Grangier, and M. Auli, "3D human pose estimation in video with temporal convolutions and semi-supervised training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7745–7754.
- [20] J. Zhang, Z. Tu, J. Yang, Y. Chen, and J. Yuan, "MixSTE: Seq2seq mixed spatio-temporal encoder for 3D human pose estimation in video," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13222–13232. [Online]. Available: <https://github.com>
- [21] W. Zhu, X. Ma, Z. Liu, L. Liu, W. Wu, and Y. Wang, "MotionBERT: A unified perspective on learning human motion representations," 2022, *arXiv:2210.06551*.
- [22] C.-J. Liang, K. M. Lundeen, W. McGee, C. C. Menassa, L. Sang-Hyun, and V. R. Kamat, "Fast dataset collection approach for articulated equipment pose estimation," in *Proc. ASCE Int. Conf. Comput. Civil Eng.*, Jun. 2019, pp. 146–152.
- [23] G. Nagymáté and R. M. Kiss, "Application of OptiTrack motion capture systems in human movement analysis," *Recent Innov. Mechatronics*, vol. 5, no. 1, pp. 1–9, Jan. 1970, doi: [10.17667/riim.2018.1/13](https://doi.org/10.17667/riim.2018.1/13).
- [24] P. Meriaux, Y. Dupuis, R. Bouteau, P. Vasseur, and X. Savatier, "A study of vicon system positioning performance," *Sensors*, vol. 17, no. 7, p. 1591, Jul. 2017, doi: [10.3390/s17071591](https://doi.org/10.3390/s17071591).
- [25] K. M. Lundeen, S. Dong, N. Fredricks, M. Akula, J. Seo, and V. R. Kamat, "Optical marker-based end effector pose estimation for articulated excavators," *Autom. Construct.*, vol. 65, pp. 51–64, May 2016, doi: [10.1016/j.autcon.2016.02.003](https://doi.org/10.1016/j.autcon.2016.02.003).
- [26] J. Kim, D. Kim, and S. H. Lee, "Construction image synthetization to overcome a small, biased real training dataset for DNN-powered visual scene understanding," in *Proc. Winter Simulation Conf. (WSC)*, Dec. 2022, pp. 2421–2428, doi: [10.1109/WSC57314.2022.10015401](https://doi.org/10.1109/WSC57314.2022.10015401).
- [27] S. Sankaranarayanan, Y. Balaji, A. Jain, S. N. Lim, and R. Chellappa, "Learning from synthetic data: Addressing domain shift for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3752–3761, doi: [10.1109/CVPR.2018.00395](https://doi.org/10.1109/CVPR.2018.00395).
- [28] J. Zhao, E. Obonyo, and Q. Yin, "Improving posture recognition among construction workers through data augmentation with generative adversarial network," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 1101, no. 9, Nov. 2022, Art. no. 092005, doi: [10.1088/1755-1315/1101/9/092005](https://doi.org/10.1088/1755-1315/1101/9/092005).
- [29] K. Stacked, G. Eilertsen, J. Unger, and C. Lundström, "Measuring domain shift for deep learning in histopathology," *IEEE J. Biomed. Health Informat.*, vol. 25, no. 2, pp. 325–336, Feb. 2021, doi: [10.1109/JBHI.2020.3032060](https://doi.org/10.1109/JBHI.2020.3032060).
- [30] H. Nguyen and H. La, "Review of deep reinforcement learning for robot manipulation," in *Proc. 3rd IEEE Int. Conf. Robotic Comput. (IRC)*, Feb. 2019, pp. 590–595, doi: [10.1109/IRC.2019.00120](https://doi.org/10.1109/IRC.2019.00120).
- [31] J. Rankin, "Reinforcement learning for trench excavation," Ph.D. dissertation, Loughborough Univ., U.K., 2023, doi: [10.26174/thesis.lboro.22714543.v1](https://doi.org/10.26174/thesis.lboro.22714543.v1).
- [32] T. Okada, T. Yamamoto, T. Doi, K. Koiwai, and K. Yamashita, "Database-driven model predictive control system for online adaptation of an autonomous excavator to environmental conditions," *Control Eng. Pract.*, vol. 145, Apr. 2024, Art. no. 105843.
- [33] R. Sekizuka, M. Ito, S. Saiki, Y. Yamazaki, and Y. Kurita, "System to evaluate the skill of operating hydraulic excavators using a remote controlled excavator and virtual reality," *Frontiers Robot. AI*, vol. 6, p. 142, Jan. 2020, doi: [10.3389/frobt.2019.00142](https://doi.org/10.3389/frobt.2019.00142).
- [34] I. Kurinov, G. Orzechowski, P. Hämmäläinen, and A. Mikkola, "Automated excavator based on reinforcement learning and multibody system dynamics," *IEEE Access*, vol. 8, pp. 213998–214006, 2020, doi: [10.1109/ACCESS.2020.3040246](https://doi.org/10.1109/ACCESS.2020.3040246).
- [35] S. Cho and S. Han, "Reinforcement learning-based simulation and automation for tower crane 3D lift planning," *Autom. Construct.*, vol. 144, Dec. 2022, Art. no. 104620, doi: [10.1016/j.autcon.2022.104620](https://doi.org/10.1016/j.autcon.2022.104620).
- [36] Q. Lu, Y. Zhu, and L. Zhang, "Excavation reinforcement learning using geometric representation," 2022, *arXiv:2201.11292*.
- [37] W. Zhao, J. P. Queralta, and T. Westerlund, "Sim-to-real transfer in deep reinforcement learning for robotics: A survey," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, Dec. 2020, pp. 737–744, doi: [10.1109/SSCI47803.2020.9308468](https://doi.org/10.1109/SSCI47803.2020.9308468).
- [38] P. Egli, D. Gaschen, S. Kersch, D. Jud, and M. Hutter, "Soil-adaptive excavation using reinforcement learning," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 9778–9785, Oct. 2022, doi: [10.1109/LRA.2022.3189834](https://doi.org/10.1109/LRA.2022.3189834).
- [39] P. Egli and M. Hutter, "A general approach for the automation of hydraulic excavator arms using reinforcement learning," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 5679–5686, Apr. 2022, doi: [10.1109/LRA.2022.3152865](https://doi.org/10.1109/LRA.2022.3152865).

- [40] L. Huang, Z. Zhu, and Z. Zou, "To imitate or not to imitate: Boosting reinforcement learning-based construction robotic control for long-horizon tasks using virtual demonstrations," *Autom. Construct.*, vol. 146, Feb. 2023, Art. no. 104691, doi: [10.1016/j.autcon.2022.104691](https://doi.org/10.1016/j.autcon.2022.104691).
- [41] Ī. Elguea-Aguinaco, A. Serrano-Muñoz, D. Chrysostomou, I. Inziarte-Hidalgo, S. Bøgh, and N. Arana-Arexolaleiba, "A review on reinforcement learning for contact-rich robotic manipulation tasks," *Robot. Comput.-Integr. Manuf.*, vol. 81, Jun. 2023, Art. no. 102517, doi: [10.1016/j.rcim.2022.102517](https://doi.org/10.1016/j.rcim.2022.102517).
- [42] Z. Jin, P. R. Pagilla, H. Maske, and G. Chowdhary, "Task learning, intent prediction, and adaptive blended shared control with application to excavators," *IEEE Trans. Control Syst. Technol.*, vol. 29, no. 1, pp. 18–28, Jan. 2021, doi: [10.1109/TCST.2019.2959536](https://doi.org/10.1109/TCST.2019.2959536).
- [43] Z. Yao, S. Zhao, X. Tan, W. Wei, and Y. Wang, "Real-time task-oriented continuous digging trajectory planning for excavator arms," *Autom. Construct.*, vol. 152, Aug. 2023, Art. no. 104916, doi: [10.1016/j.autcon.2023.104916](https://doi.org/10.1016/j.autcon.2023.104916).
- [44] M. Allain, S. Konduri, H. Maske, P. R. Pagilla, and G. Chowdhary, "Blended shared control of a hydraulic excavator," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 14928–14933, Jul. 2017, doi: [10.1016/j.ifacol.2017.08.2541](https://doi.org/10.1016/j.ifacol.2017.08.2541).
- [45] CAT 320 Hydraulic Excavator. Accessed: Apr. 5, 2022. [Online]. Available: <https://magomhrc.com/en/catalogue/205-cat-320-hydraulic-excavator-with-option-parts.html>
- [46] R12DS. Accessed: Oct. 10, 2022. [Online]. Available: <https://www.radiolink.com/r12ds>
- [47] M. Scarpino, *Motors for Makers: A Guide to Steppers, Servos, and Other Electrical Machines*. Seattle, WA, USA: Que Publishinh, 2015.
- [48] J.-D. Warren, J. Adams, and H. Molle, "Arduino for Robotics," in *Arduino Robotics*. Berkeley, CA, USA: Apress, 2011, pp. 51–82, doi: [10.1007/978-1-4302-3184-4_2](https://doi.org/10.1007/978-1-4302-3184-4_2).
- [49] *Arduino Mega 2560*. [Online]. Available: <https://store-usa.arduino.cc/products/arduino-mega-2560-rev3?selectedStore=us>
- [50] HC-05 Bluetooth Module. Accessed: Jun. 10, 2022. [Online]. Available: <chrome-extension://efaidnbmnnnnibpcajpgclclefindmkaj/https://cdn.awsli.com.br/945/945993/arquivos/HC-05-BOARD.pdf>
- [51] I. Pardoe, *Applied Regression Modeling*. Hoboken, NJ, USA: Wiley, 2020, doi: [10.1002/9781119615941](https://doi.org/10.1002/9781119615941).
- [52] J. Xu and H.-S. Yoon, "A review on mechanical and hydraulic system modeling of excavator manipulator system," *J. Construction Eng.*, vol. 2016, pp. 1–11, Oct. 2016, doi: [10.1155/2016/9409370](https://doi.org/10.1155/2016/9409370).
- [53] C. Finn, T. Yu, J. Fu, P. Abbeel, and S. Levine, "Generalizing skills with semi-supervised reinforcement learning," 2016, *arXiv:1612.00429*.
- [54] Y. Song, Y. Zhou, A. Sekhari, J. A. Bagnell, A. Krishnamurthy, and W. Sun, "Hybrid RL: Using both offline and online data can make RL efficient," 2022, *arXiv:2210.06718*.
- [55] *Built Robotics Exosystem*. Accessed: Nov. 15, 2022. [Online]. Available: <https://www.builtrobotics.com/technology/exosystem>



SEYEDEH FATEMEH SAFFARI received the B.Sc. degree in engineering science from the University of Tehran, in 2019. She is currently pursuing a Ph.D. degree at the University of Toronto. Her research focuses on generating synthetic datasets of construction machinery in controlled settings to advance digital twinning and automation in the construction industry. She aims to contribute to creating safer and more productive work environments through her research.



His research interests include construction automation, robotics, and digitization, particularly addressing data scarcity in construction AI training.



DAEHO KIM received the bachelor's and master's degrees in architectural engineering from Yonsei University, South Korea, the master's degree in construction engineering and management from the University of Michigan, as a John L. Tishman Master's Fellowship Recipient, and the Ph.D. degree in civil engineering (construction engineering and management) from the University of Michigan. He also received the Graduate Certificate in computational discovery and engineering.

BYUNGJOO CHOI received the Ph.D. degree in civil engineering from the University of Michigan. He also received the Graduate Certificate in complex systems from the University of Michigan. His research interests include enhancing construction performance through the integration of advanced technologies such as artificial intelligence, robotics, and construction automation. His interests also extend to the innovation of construction production systems through the application of off-site construction methodologies.

...